STRUCTURAL AND MECHANISTIC INSIGHTS INTO *STREPTOCOCCUS PNEUMONIAE* NADPH OXIDASE

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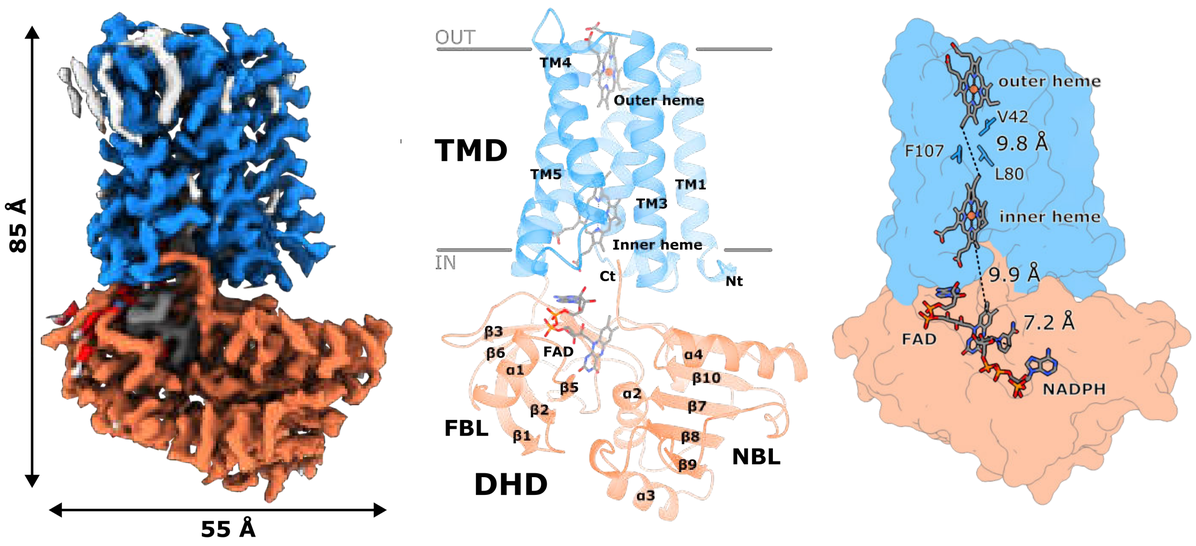
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Background

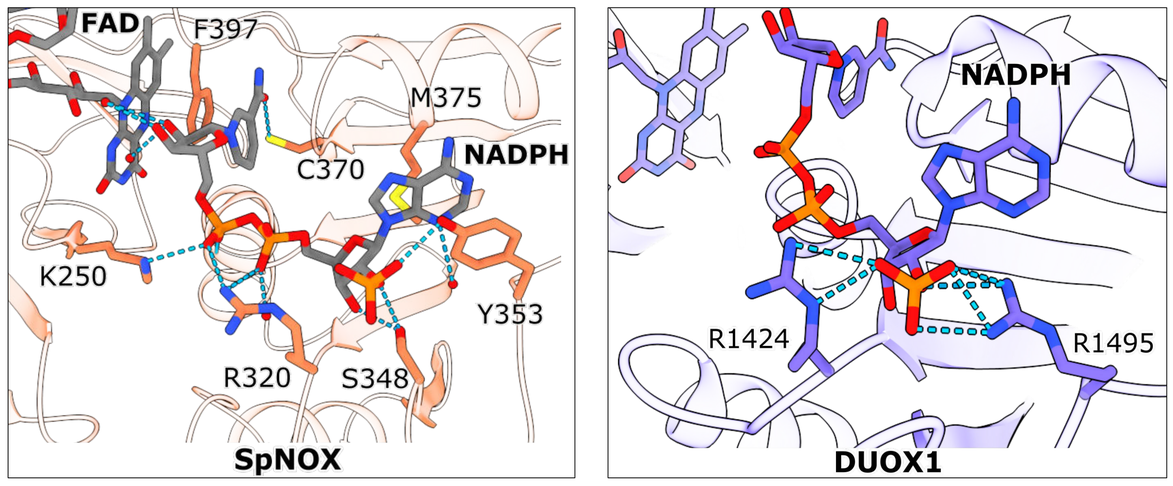
The NADPH oxidase (NOX) protein family consists of 7 members in humans. These transmembrane proteins exclusively produce reactive oxygen species (ROS) and have been shown to be important for immunity, cell signalling and cancer [1]. Structural studies of NOX proteins have been proven to be difficult due to low yield from purifications and low stability. While the NOXs were initially thought to be exclusive to Eukaryotes, proteins sharing the catalytic core were found in Bacteria [2]. The best characterized member is *Streptococcus pneumoniae* NADPH oxidase (SpNOX). SpNOX is a 46-kDa membrane protein which is readily purified and shows high stability and activity in detergent micelles. SpNOX can thus be used as a model of NADPH oxidases in order to gain structural and mechanistic insights.

Cryo-EM structure of SpNOX



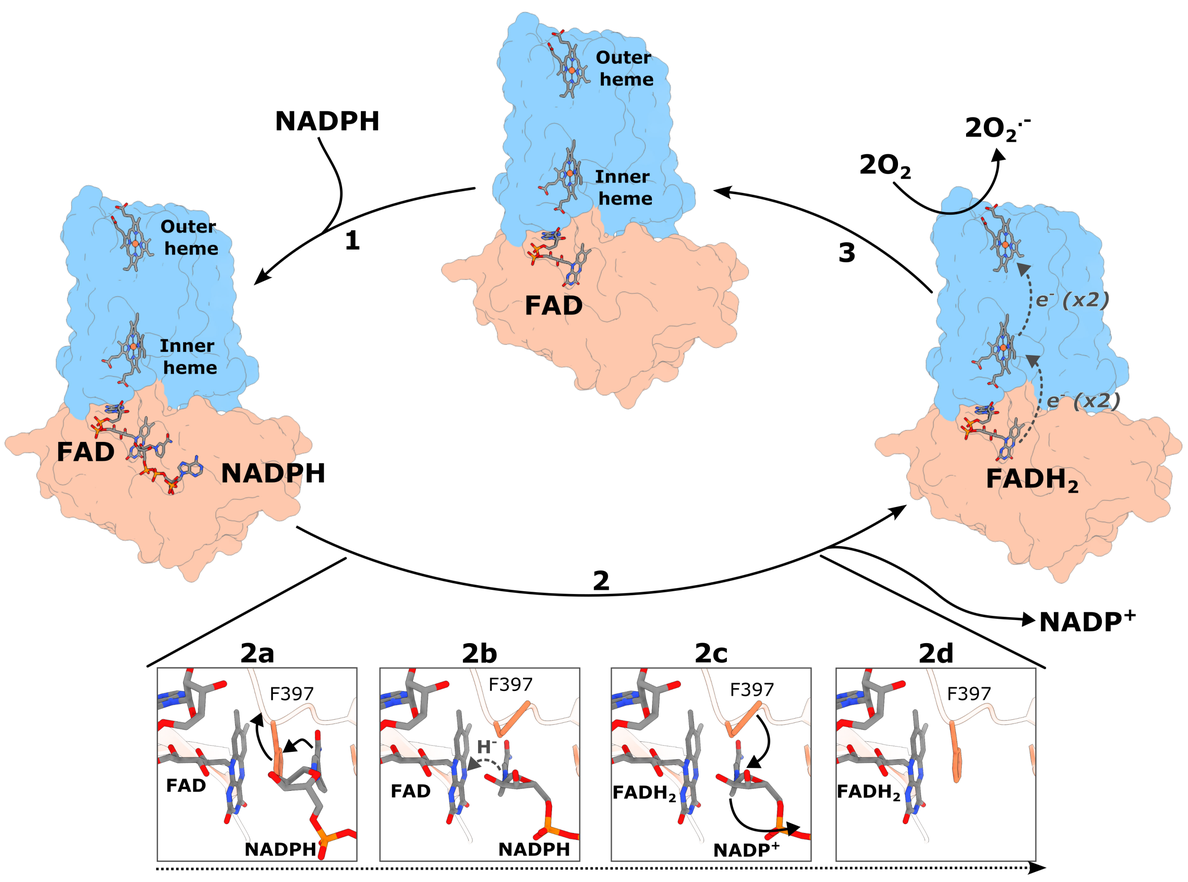
Four structures of SpNOX were determined with single-particle analysis cryo-EM (2.2-2.6 Å). The transmembrane domain (TMD) contains six helices which sandwich two hemes in the lipid bilayer. The soluble dehydrogenase domain (DHD) consists of two lobes, a flavin binding lobe (FBL) and a nucleotide binding lobe (NBL) which bind an FAD and NAD(P)H, respectively. All redox cofactors are within efficient electron transfer distance (<14 Å) from each other in all four structures.

Substrate specificity for NOXs



Eukaryotic NOX proteins exclusively use NADPH as substrate, while SpNOX uses both, NADH and NADPH, with similar efficiency. This can be explained because SpNOX has a tyrosine (Y353) replacing an arginine present in all mammalian NOXs, like R1495 in DUOX1 (PDB: 7D3F). This arginine interacts with the 2'-phosphate of NADPH, conferring substrate selectivity.

Proposed catalytic cycle of SpNOX



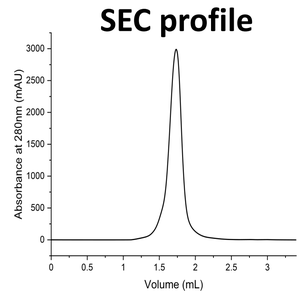
With our structures we were able to propose a catalytic cycle for SpNOX. **1,** NADPH binds SpNOX in the resting state. **2a,** Phe397 is transiently displaced by NADPH. **2b,** NADPH approaches FAD and transfers a hydride. **2c,** Phe397 displaces $NADP^+$ and the substrate is released while the protein returns to the inactive Phe397 conformation (**2d**). **3,** $FADH_2$ can transfer two electrons to the transmembrane hemes. The outer heme reduces two O_2 generating two $O_2^{\cdot-}$ which returns the protein to the resting state.

Cryo-EM of 46-kDa SpNOX

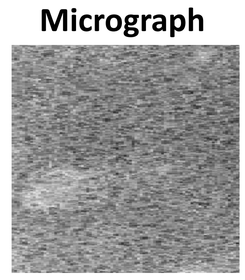
Krios G4, Falcon IV, Selectris X



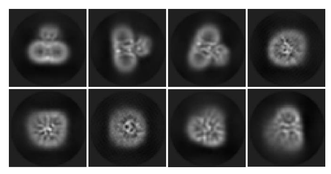
SEC profile



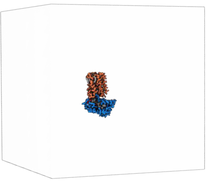
Micrograph



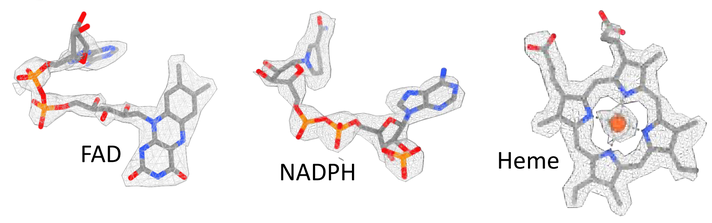
2D classes



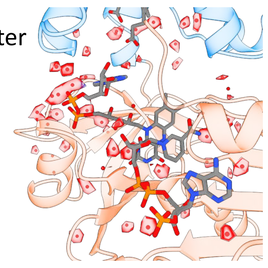
Final extraction box



High-resolution features

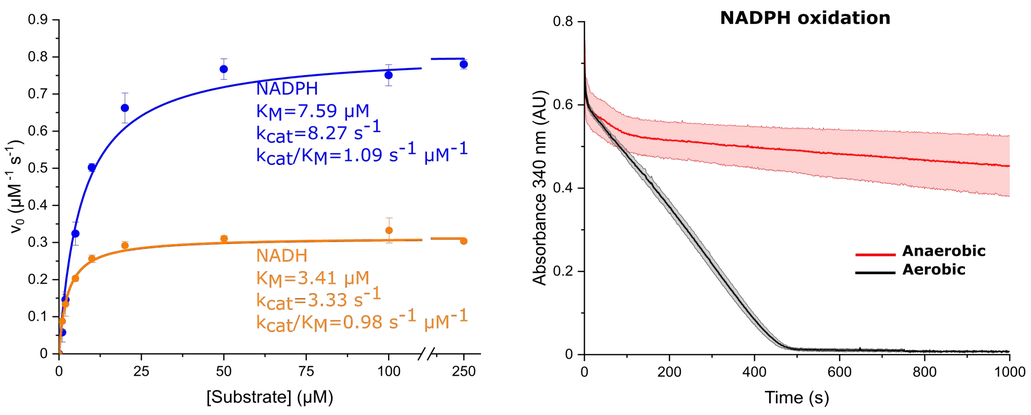


Resolved water molecules



A highly-optimized sample and the latest electron microscope hardware were used for cryo-EM. Reaching high resolution (up to 2.2 Å) was achieved in part due to a few key processing steps related to particle picking, box size and mask size aside from optimal biochemical conditions.

NOX-homologous activity of SpNOX



SpNOX shows activity with NADH and NADPH as substrates. In contrast, mammalian NOXs use NADPH exclusively. Moreover, it exhibits oxygen-dependent NADPH oxidation, which indicates that it can function homologously to the human NOX proteins. Nevertheless, the physiological function of SpNOX is still unknown.

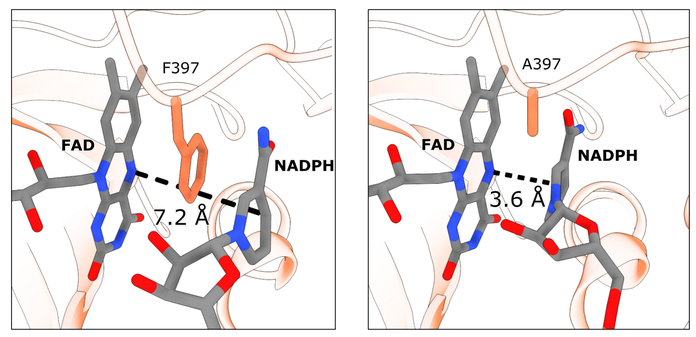
F397A mutant in the active state

SpNOX C-terminal sequence

390 395 400

...QNPKT ELIYE GFKFK*

...QNPKT ELIYE GAKFK*



The SpNOX structure frozen under turnover conditions did not show the active state of the protein. A C-terminal phenylalanine (Phe397) blocks the approach of the NADPH to the FAD. A Phe397Ala mutant showed higher catalytic activity and allowed us to capture the productive state with cryo-EM.

Summary

We were able to resolve high-resolution cryo-EM structures (2.2-2.6 Å) of a 46-kDa membrane protein without the help of fiducials. This was achieved due to a highly optimized sample paired with state-of-the-art microscope hardware. With this, we are able to resolve the first structure of a bacterial NOX protein and we shed light onto the catalytic cycle and substrate binding of NOXs.

References & Acknowledgements

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Out on NSMB!

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